Assignment 1 Network Analysis

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Abstract

This study provides a comprehensive network analysis of the global air transportation system, using a dataset where nodes represent airports and edges denote the flight routes between them. The research employs key metrics such as average path length, network density, and the size of the largest connected component to describe the network's topology and functionality. The findings indicate efficient global connectivity, with an average path length of 4.103 and a network density of 0.0033, highlighting the strategic role of major hub airports. This analysis identifies critical nodes and offers insights into operational resilience and strategic planning, which can guide targeted infrastructure improvements and policy decisions. Overall, the paper contributes to the understanding of air transport networks as complex systems, enhancing network management and development for global air travel.

1. Introduction

In the interconnected world of global travel, airports serve as central nodes in the extensive air transport network. The structure and efficiency of this network are crucial for both local and international connectivity, influencing economic, social, and environmental dynamics. This paper presents a comprehensive network analysis of a dataset where nodes represent airports and edges correspond to the routes between them. Through this analysis, we aim to uncover the underlying patterns and characteristics of the global air transport network.

A variety of network metrics will be employed in order to elucidate the structure and connectivity of the network. Metrics such as `degree centrality`, `betweenness centrality` and `closeness centrality` are calculated to identify key airports and understand their role within the network. In addition, we examine the `modularity` of the network to determine the presence of community structures, which may reflect geopolitical regions or airline alliances.

By analysing these metrics, we aim to draw conclusions about the resilience of the air transport network, identify critical hubs essential for maintaining global connectivity, and suggest potential areas for improving network efficiency. The results of this study could be intended to contribute to the strategic planning and development of more robust, efficient and sustainable air transport networks.

2. Dataset

I personally analysed a real graph of large size, created thanks to the Routes.csv dataset, where we connect different airports thanks to the columns `source` and `destination airports`.

Figure 2.1: Graph built from the Routes.csv dataset

The network of global air transportation, as modeled through our dataset, comprises 3425 airports (nodes) interconnected by various direct routes (edges). This vast network facilitates a detailed examination of several critical network properties that elucidate the overall connectivity and efficiency of the system. Since the resulting graph was not connected, I looked for the largest connected component and discarded the rest (28 nodes). Then I computed the Average Path Length and the density of the graph.

3. Analysis of Network Metrics

The following metrics have been calculated in order to assess the structure and performance of this network.

3.1. Average Path Length

The average path length in the network is approximately 4.103. This metric indicates that, on average, any airport can connect to another within approximately four routes. A shorter average path length in a network of this size is indicative of the efficiency of global air travel, facilitating rapid and direct connections across vast geographical distances.

3.2. Density

The network's density is 0.0033, which indicates that the network is extensive but sparse. This is typical for transport networks where direct routes between every pair of nodes (airports) are neither practical nor economically viable. The sparse nature of the network emphasises the strategic importance of hub airports that connect indirect routes.

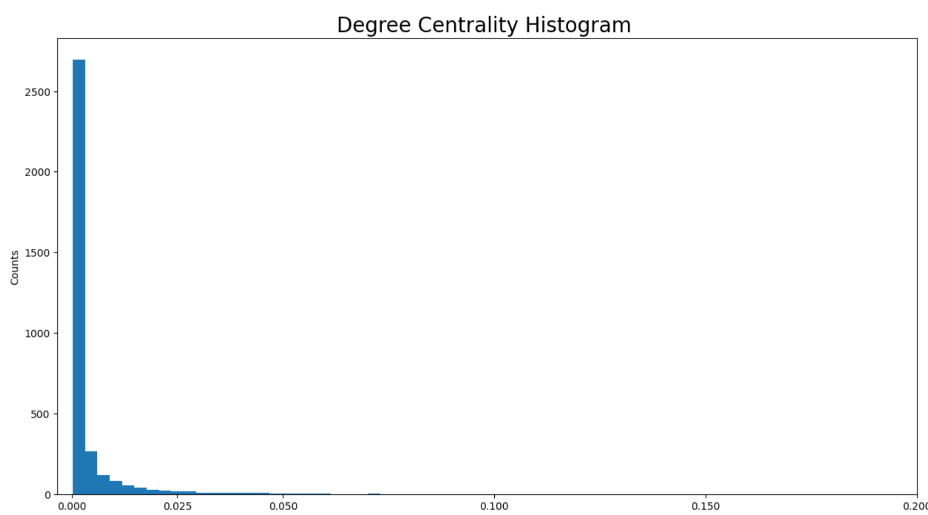
3.3. Size of the Largest Connected Component (LCC)

The largest connected component of the graph comprises 3397 nodes and 19231 edges. This component constitutes the overwhelming majority of the network, indicating that nearly the entire network is reachable. This reinforces the global nature of air transportation, where almost any airport can be accessed from any other in the network via successive flights.

The analysis shows a network with good connections between airports, even though it is not very dense. This makes it easy to get from one place to another and to cover a large area.

3.4. Degree Centrality

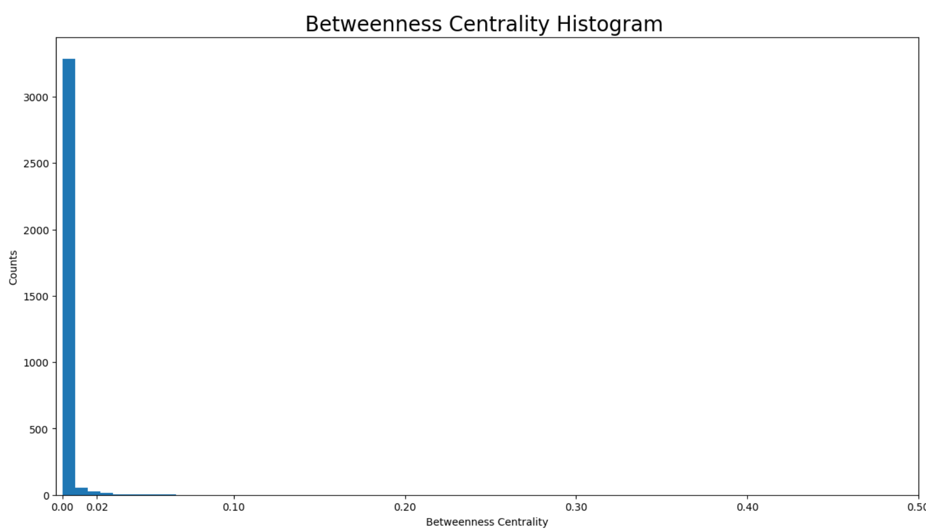
Degree centrality is a measure of how connected a node is. The more connected a node is, the more edges it has and the more neighbour nodes it has. In fact, the degree of centrality of a node is the fraction of nodes it is connected to. It is the percentage of the network that the node is connected to.



The histogram indicate a highly skewed distribution, where a majority of airports have a low number of direct connections, typically fewer than 20 routes. This is visible from the tall bar at the left side of the histogram. As the number of routes increases, the number of airports that have such a high number of connections decreases significantly, which is evidenced by the long tail extending towards the right. This pattern is typical of scale-free networks, which are distinguished by the presence of a few highly connected hubs (airports with a high degree of connectivity), while the majority of nodes (airports) have relatively few connections. These hub airports are of crucial importance for the network's connectivity, facilitating efficient global travel by linking many less connected airports to the broader network.

3.5. Betweenness Centrality

Betweenness centrality is a key metric in network analysis that measures the extent to which a node lies on the shortest paths between other nodes in the network. In the context of an air transportation network, where nodes represent airports and edges represent direct flight routes, betweenness centrality identifies which airports act as critical connectors or hubs in the network.

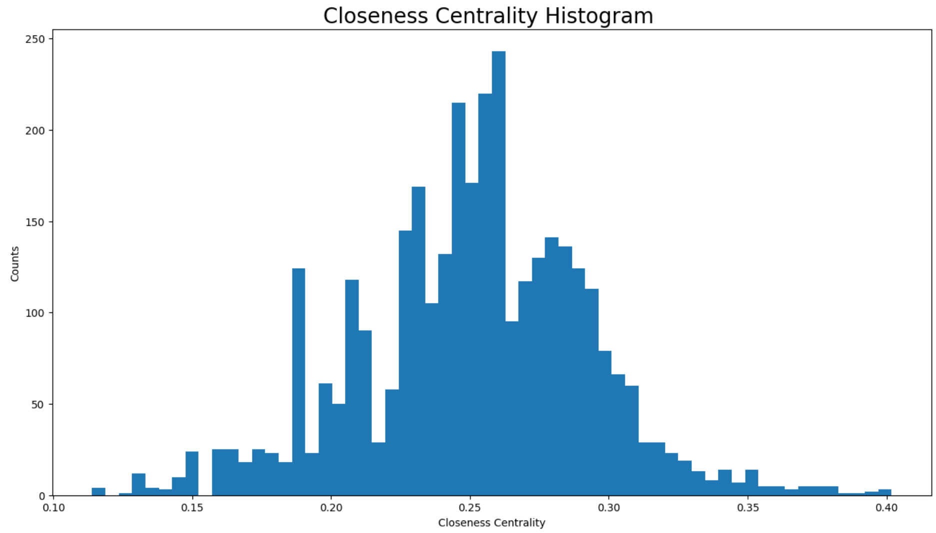


A histogram of betweenness centrality values reveals that the majority of airports exhibit a low betweenness centrality, as evidenced by the high bar at the near-zero values of the x-axis. This suggests that the majority of airports do not act as significant intermediaries in the flow of air traffic between other airports. In other words, these airports are either end nodes or are bypassed by more efficient or direct routes.

This distribution serves to highlight the existence of a limited number of critical nodes in the network whose operational capacity and security are of vital importance for the continued functionality and robustness of global air transportation.

3.6. Closeness Centrality

The Closeness Centrality, which has been analysed in the context of airports and routes, provides insights into the global accessibility of each airport.



The Closeness Centrality Histogram demonstrates a multi-modal distribution, indicating that airports cluster into groups with similar accessibility levels. The majority of airports exhibit closeness centrality values between 0.15 and 0.35, indicating a range of accessibility levels within the network. Airports with higher closeness centrality are of significant importance, as they facilitate more direct and expedient connections to a multitude of other airports. This enhances the network's overall efficiency and reduces travel times. Conversely, airports with lower closeness values are less central, often requiring more transfers and longer routes, which highlights potential for improving their connectivity in order to enhance network resilience and efficiency.

4. Most important nodes

A review of the table of the best airports for Degree, Betweenness and Closeness we can derive some interesting insights:

- Overlap in Centrality Measures: Charles-de-Gaulle (CDG) and Frankfurt (FRA) appear across all three centrality measures, indicating their comprehensive strategic importance to the network. They serve not only as major hubs with numerous direct connections but also as critical transit points in the paths between numerous other airports, and they can quickly reach a large number of nodes.
- Regional Hubs: Different airports serve as regional powerhouses like IST in Istanbul for connecting to various parts of Europe and the Middle East, and LAX (Los Angeles) for connections across the Pacific and within the Americas.

Degree		Betweenness		Closeness	
Node	**Value**	**Node**	**Value**	**Node**	**Value**
AMS Amsterdam	0.0730	ANC Ted Stevens	0.0736	FRA Frankfurt	0.4018
FRA Frankfurt	0.0718	LAX Los Angeles	0.0662	CDG Charles-de-Gaulle	0.3993
CDG Charles-de-Gaulle	0.0706	CDG Charles-de-Gaulle	0.0629	LHR Heathrow	0.3973
IST Istanbul	0.0694	DXB Dubai	0.0574	AMS Amsterdam	0.3930
ATL Atlanta	0.0638	FRA Frankfurt	0.0519	DXB Dubai	0.3926

Table 4.1: Network Metrics for Nodes

- Special Mention of Anchorage (ANC): Although not typically seen as a global hub like others, its high betweenness centrality suggests a significant role in trans-pacific routes, acting as an efficient transit point between North America and Asia.

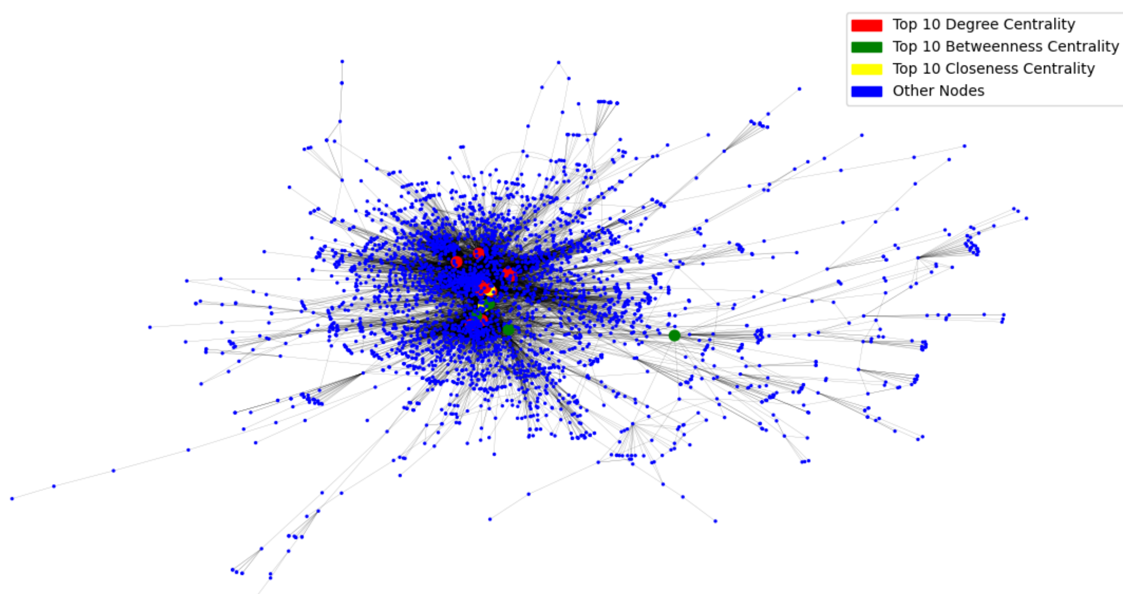
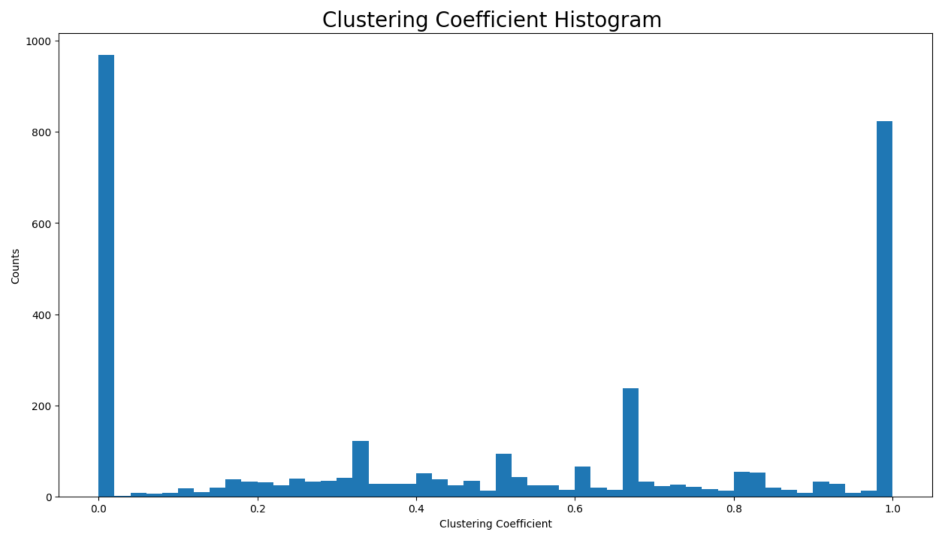


Figure 4.1: Top 10 nodes for each metric

Furthermore, it can be observed that the strategic nodes of the network are positioned in the centre of the graph, in accordance with expectations. This is where all the hubs are situated.

5. Clustering Effects

The clustering coefficient in our field is a measure of the extent to which airports form tightly knit groups, where direct routes exist between a specific airport and others, as well as between those other airports themselves.



The clustering coefficient histogram of the airport-route network displays a bimodal distribution, which highlights two predominant patterns. Firstly, airports either do not participate in any triangular connections, indicating isolation or specialised routes. Secondly, they are part of highly interconnected clusters, which are likely to represent regional hubs or strong domestic networks. The absence of airports with moderate clustering coefficients indicates a clear division between highly integrated and minimally connected airports.

This stark contrast in clustering dynamics suggests different operational roles within the network. Those airports with high clustering coefficients serve as crucial nodes for regional connectivity, facilitating efficient local travel. In contrast, those with low clustering coefficients might focus on specific, possibly longer-distance routes that do not form triangles with other routes. From a strategic perspective, this insight could inform improvements in connectivity for isolated airports and strengthen the infrastructure in clustered areas, thereby enhancing overall network resilience and efficiency.

6. Bridges

In the previous sections, it was demonstrated that there are different types of nodes, namely those that function as hubs and the others. It is of paramount importance to study the bridges within an airport-route network in order to gain a deeper understanding of the resilience and robustness of the entire air transportation system. Bridges in a network graph are edges whose removal would result in an increase in the number of disconnected subgraphs, or components, thereby fragmenting the network. In our context, bridges represent critical routes that, in the event of disruption, could result in the isolation of specific airports or regions, thereby significantly impacting connectivity and efficiency.

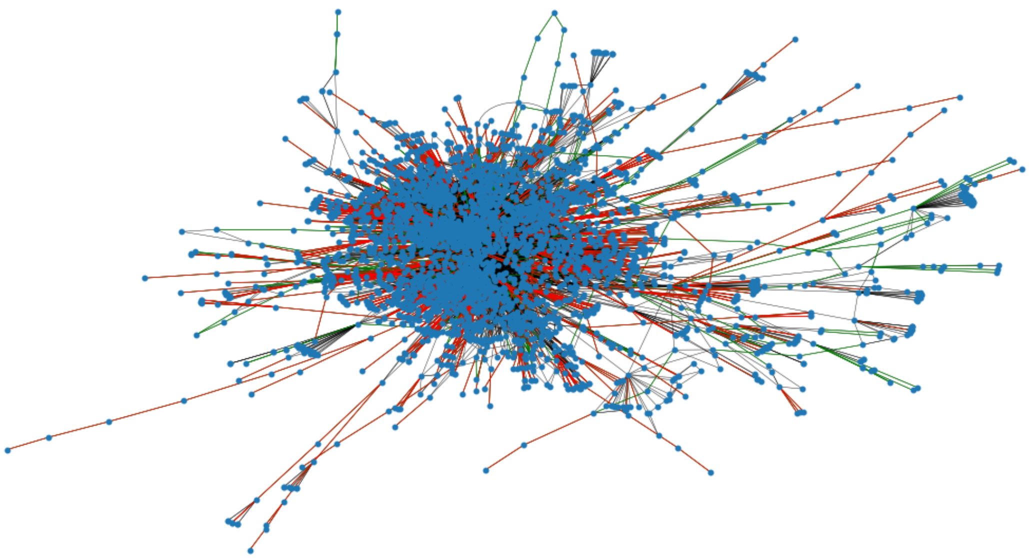


Figure 6.1: The bridges can be seen with the red color and the local bridges with the green color. Black edges are neither local bridges nor bridges.

The presence of 805 bridges in a network with 19,231 edges suggests that a relatively small but significant portion of the edges (about 4.2%) are critical for maintaining the connectivity of the network. Removing these edges could potentially fragment the network significantly, affecting its overall robustness and functionality.

Having 1,272 local bridges indicates that there are many edges which, although not critical for connectivity (as in the case of bridges), still do not participate in closed triangles. This suggests that these edges might be connecting different clusters or communities within the network, serving as "weak ties."

The majority of black edges, which are not bridges or local bridges, can be found in the central conglomeration, where the dense cluster of nodes is likely to represent the major hub airports. The aforementioned hubs are highly connected, indicating that they serve as central points for routing traffic between various other, less connected airports.

7. Network Communities

A community is defined as a group of nodes, with each node within the group connected to a greater number of other nodes than between groups. An asynchronous algorithm (community.asynfluidc) will be employed for the detection of communities within this network.

Figure 7.1: communities graph

The graph divides the global air transport network into eight distinct communities to explore geographic and strategic connections. This division enhances understanding of the network's operational dynamics and resilience, aiding in route optimization and guiding targeted infrastructure investments. Each community shows dense internal connections, highlighting the modular nature of the network and facilitating focused management and policy development across these clusters.

8. Conclusion

This network analysis of the global air transportation system, utilising centrality metrics, has provided deep insights into the structure and efficiency of the network. The analysis revealed that airports such as Charles de Gaulle, Frankfurt, and Amsterdam serve as critical nodes within the global air transport network. This is not only due to their high number of direct connections, but also because of their strategic positions in facilitating efficient pathways across the network. These hubs enhance connectivity, making global travel more accessible and efficient.

The centrality metrics, namely degree, betweenness, and closeness, each highlighted different aspects of the network's functionality. Degree centrality identified the most connected airports, betweenness centrality pointed out those that are most crucial for network flow and resilience, and closeness centrality illuminated the airports that are most efficiently positioned to access the entire network. This analysis underscores the importance of strategic planning and management of these key nodes to ensure network resilience and efficiency. Furthermore, the study highlights the necessity for continued investment in these hubs to enhance capacity and efficiency, given the continued growth of global air traffic. The findings of the study could be of significant importance for policymakers, airport authorities, and airline companies aiming to optimise operations and develop strategies for future expansion and crisis management. In conclusion, this comprehensive network analysis not only illustrates the current state of global air traffic networks but also provides a roadmap for enhancing and sustaining the essential infrastructure required for the continued growth and robustness of global connectivity.